Akoizo Original Practice - Form 1 - Answer key

Division C | Protein Modeling | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) A PDB identifier alone fixes the chosen chain?

Unit: Event evidence, reference design, and model purpose.
Worked-case data: A structure entry has chains A and B and a gap at residues 40-45. Plan a model label.

A. No

B. Always

C. Only for viruses

D. Only for enzymes

Correct: A

2. (2 points) Hydrophobic attraction is a new covalent bond?

Unit: Amino-acid chemistry, folding, and stability.
Worked-case data: A buried leucine is replaced by a charged residue. Predict cautiously.

A. No

B. Always

C. Only between leucines

D. Only in water

Correct: A

3. (2 points) Rotation changes…

Unit: Molecular viewing and reproducible JUDE/Jmol workflows.
Worked-case data: Plan a figure highlighting one chain’s helix region.

A. Viewpoint

B. Primary sequence

C. Chemical bonds automatically

D. PDB identifier

Correct: A

4. (2 points) A stop codon ordinarily specifies…

Unit: Protein synthesis and diverse functions.
Worked-case data: Coding DNA is 5′-ATG GAA TAA-3′. Give simple unprocessed mRNA and translated product.

A. One universal amino acid

B. A glycan

C. A DNA base

D. Termination

Correct: D

5. (2 points) An epitope is…

Unit: Viruses, entry, and immune recognition.
Worked-case data: An antibody binds a viral protein in an assay. Does that prove neutralization?

A. The entire immune system

B. Every viral genome

C. A unit of salinity

D. A recognized molecular feature

Correct: D

6. (2 points) A missing segment should be…

Unit: Pre-build planning, annotation, and model review.
Worked-case data: At a scale of 1 cm per Å, how long is a modeled 12 Å separation?

A. Silently invented

B. Called observed

C. Ignored in numbering

D. Explicitly marked or qualified

Correct: D

7. (2 points) Denaturation always cleaves peptide bonds?

Unit: Structural levels and denaturation.
Worked-case data: Heating disrupts activity but the amino-acid sequence remains intact. What changed?

A. Only for helices

B. Only at neutral pH

C. No

D. Always

Correct: C

8. (2 points) The 2009 outbreak is used here as…

A. A permanent event assignment

B. A diagnosis guide

C. A historical case study

D. A current case count

Correct: C

9. (4 points) Why verify current requirements?

Unit: Event evidence, reference design, and model purpose.
Worked-case data: A structure entry has chains A and B and a gap at residues 40-45. Plan a model label.

answer: Assigned proteins and construction constraints can differ from a supplied course case study. (4 points).

10. (4 points) Why does pH matter?

Unit: Amino-acid chemistry, folding, and stability.
Worked-case data: A buried leucine is replaced by a charged residue. Predict cautiously.

answer: It changes protonation and interactions, often in context-dependent ways. (4 points).

11. (4 points) Why reset a selection?

Unit: Molecular viewing and reproducible JUDE/Jmol workflows.
Worked-case data: Plan a figure highlighting one chain’s helix region.

answer: Later operations might otherwise affect only an unintended subset. (4 points).

12. (4 points) What does a tRNA do?

Unit: Protein synthesis and diverse functions.
Worked-case data: Coding DNA is 5′-ATG GAA TAA-3′. Give simple unprocessed mRNA and translated product.

answer: It links codon recognition with delivery of its attached amino acid. (4 points).

13. (4 points) Why inspect conformational states?

Unit: Viruses, entry, and immune recognition.
Worked-case data: An antibody binds a viral protein in an assay. Does that prove neutralization?

answer: Entry proteins can rearrange during function. (4 points).

14. (4 points) What can peer tracing reveal?

Unit: Pre-build planning, annotation, and model review.
Worked-case data: At a scale of 1 cm per Å, how long is a modeled 12 Å separation?

answer: Ambiguous path, labels, orientation, or hidden structural features. (4 points).

15. (4 points) Why is a loop not meaningless?

Unit: Structural levels and denaturation.
Worked-case data: Heating disrupts activity but the amino-acid sequence remains intact. What changed?

answer: Loops can form binding sites, switches, or connectors. (4 points).

16. (4 points) Distinguish drift and reassortment.

Unit: Influenza H1N1 and hemagglutinin.
Worked-case data: A model shows one HA chain. What must be checked before describing the functional surface?

answer: Drift accumulates changes; reassortment exchanges whole segments. (4 points).

17. (4 points) What does the synthetic lab not prove?

Unit: Event evidence, reference design, and model purpose.
Worked-case data: A structure entry has chains A and B and a gap at residues 40-45. Plan a model label.

answer: An actual protein’s conformation, energy minimum, or biological function. (4 points).

18. (4 points) What is missing from the lab’s contact count?

Unit: Amino-acid chemistry, folding, and stability.
Worked-case data: A buried leucine is replaced by a charged residue. Predict cautiously.

answer: Atom identities, solvent, charge states, bond geometry, and an energy model. (4 points).